FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – XI

PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**ABC**	19.	**AC**	37.	**ABD**
2.	**D**	20.	**AB**	38.	**BC**
3.	**ABCD**	21.	**BCD**	39.	**BCD**
4.	**A**	22.	**ACD**	40.	**AB**
5.	**ABC**	23.	**AC**	41.	**D**
6.	**AC**	24.	**BCD**	42.	**BC**
7.	**AD**	25.	**ACD**	43.	**B**
8.	**C**	26.	**B**	44.	**D**
9.	**A**	27.	**B**	45.	**A**
10.	**D**	28.	**C**	46.	**B**
11.	**B**	29.	**B**	47.	**C**
12.	**A**	30.	**C**	48.	**D**
13.	**C**	31.	**D**	49.	**D**
14.	**3**	32.	**2**	50.	**5**
15.	**9**	33.	**5**	51.	**2**
16.	**4**	34.	**4**	52.	**3**
17.	**5**	35.	**4**	53.	**8**
18.	**2**	36.	**6**	54.	**7**

Physics

PART – I

SECTION – A

1. **ABC**

$$u = \frac{kx}{(x^2+y^2)^{3/2}}$$

$$F_x = -\left(\frac{\partial u}{\partial x}\right)_{(y-CON.)}$$

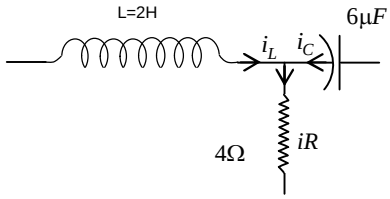
$$F_y = -\left(\frac{\partial u}{\partial y}\right)_{(x-CON.)}$$

After solving find $\bar{a}_t$ and $\bar{a}_r$ *etc*

2. **D**

Use $i_L + i_C = i_R; v_L = \frac{Ldi_L}{dt}$ etc.

$$q = CV = \frac{dq}{dt} = C\left(\frac{dV}{dt}\right) + V\left(\frac{dC}{dt}\right)$$



3. **ABCD**

$$i_1 = \frac{\varepsilon_0}{\sqrt{(R^2 + X_c{}^2)}}\sin(\omega t + 53^o) = 4\sin(\omega t + 53^o)$$

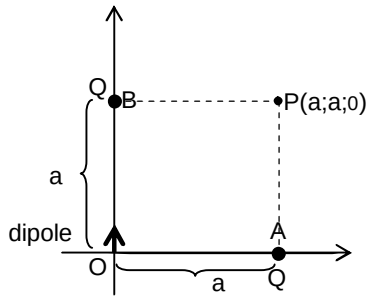
$$i_2 = \frac{\varepsilon_0}{\sqrt{(R^2 + X_L{}^2)}}\sin(\omega t - 37^o) = 4\sin(\omega t - 37^o)$$

Voltmeter & ammeter reads rms value

4. **A**

$$\overrightarrow{E_P} = \overrightarrow{E_A} + \overrightarrow{E_B} + \vec{E}_{dipole}$$

$$Finally\ |\overrightarrow{E_P}| = \frac{Q}{4\pi \in_0 a^2\sqrt{2}}(4^2+1^2)^{1/2} = \sqrt{\frac{17}{2}}\frac{Q}{4\pi \in_0 a^2}$$



5. **ABC**

$$A = A_0 e^{-\lambda t_1} \qquad -(i)$$

$$2A/3 = A_0 e^{-\lambda t_2} \qquad -(ii)$$

$$T_{mean} = 1/\lambda = \frac{(t_2 - t_1)}{\ell n_e(3/2)}$$

$$T_{1/2} = \frac{\ell n_e 2}{\lambda}$$

probability of survival of a nucleus = e^{-1}

6. **AC**

$$Y = \frac{stress}{strain} = \frac{T/S}{\Delta L/L} = \frac{TL}{S\Delta L}$$

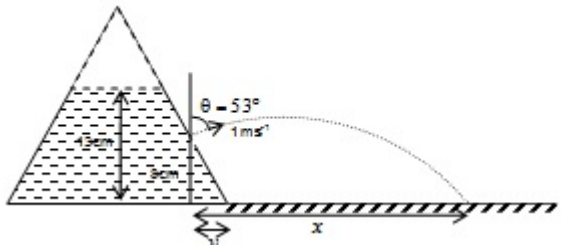
$$T = SY\frac{\Delta L}{L}$$

$$\nu_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

$$\nu_1 = 200Hz$$

Hence, the fourth harmonic (or 3rd overtone) has a frequency of 800 Hz.

7. **AD**



$$8/y = \tan 53^o = \frac{4}{3}$$

$$v = \sqrt{2g(\Delta n)} = 1ms^{-1} \text{ to get 't'}$$

$$8/100 = \frac{3t}{5} - 5t^2 \Rightarrow t = (1/5)\sec$$

$$So \quad x = \frac{1\times 1}{5}\times\frac{4}{5}\times 100 = 16\,\text{cm}$$

$$16 - 6 = 10\,cm$$

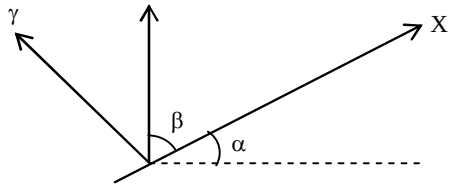
$$F_H = mv\cos 37^o = \rho A v^2 \times \frac{4}{5} = 0.12 \text{ Newton}$$

8. **C**
9. **A**
10. **D**

Sol. **(for 8-10).** After calculating total time of flight $(O \longrightarrow C)$

$$T_0 = \frac{V_0\cos\beta}{q\sin\alpha}$$

$to_A = \frac{2V_0\sin\beta}{q\cos\alpha}$; Rebound velocity at $A \longrightarrow (eV_0\sin\beta)$



Calculate t_{AB} & t_{BC}

Using To = $t_{OA} + t_{AB} + t_{BC}$ we get $\beta = \cot^{-1}\left(\frac{13}{4}\tan\alpha\right)$

$$K = \frac{V_0(\cot\beta)}{(q\sin\alpha)} = \frac{(\cot\beta)}{(\sin\alpha)}(V_0 = 10ms^{-1})$$

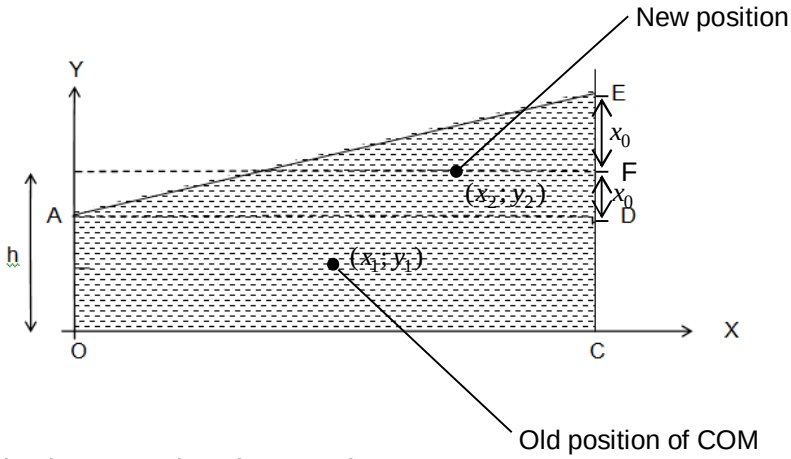
11. **B**
12. **A**
13. **C**

Sol. **(for 11-13).** use I[st] law of thermodynamics and apply for different given process.

SECTION – C

14. **3**

The liquid surface after displacing from eq. position



Write down equation of centre of mass

$$(y_2 - y_1) = (y - y_1) = \frac{x_0^2}{6h}$$

$$(x_2 - x_1) = (x - x_1) = \frac{Lx_0}{3h}$$

Hence

$$y - y_1 = \frac{3h}{2L^2}(x - x_1)^2$$

Restoring force Ma = $-\frac{3}{2}\frac{mgh \times 2X}{L^2}$

$$a = -\frac{3gh \times X}{L^2}$$

Finally $\omega^2 = \frac{3gh}{L^2}$

15. **9**

$$m\left\{r\frac{d^2\theta}{dt^2} + 2\left(\frac{dr}{dt}\right)\left(\frac{d\theta}{dt}\right)\right\} = eB\left(\frac{dr}{dt}\right)$$

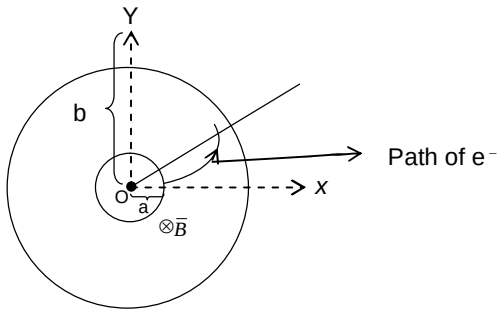
$$\Rightarrow \frac{d}{dt}(r^2\dot{\theta}) = \frac{eB}{m}r\dot{r}$$

$$\frac{d}{dt}(r^2\dot{\theta}) = \frac{eB}{2m}\left(\frac{d}{dt}r^2\right)$$

$$\Rightarrow r^2\dot{\theta} = \left(\frac{eB}{2m}\right)(r^2 - a^2) \qquad \left\{\frac{d\theta}{dt} = 0 \text{ for r = a}\right\}$$

& $\dot{\theta} = \frac{eB}{2m}\left(1 - a^2/b^2\right)$

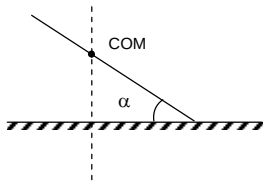
$$v^2 = \frac{2eV}{m} = \left(\frac{e^2B^2}{4m^2}\right)b^2(1 - a^2/b^2)$$



Put the value & get the result.

16. **4**

Sol.



COM remains on this line

17. **5**

$$\Delta E = 3.4eV - \left(-13.6\frac{z^2}{n^2}\right)$$

$$= 17eV$$

$$\frac{hc}{\lambda_1} = 17eV \quad \text{_____(1)}$$

2nd condition $\quad \Delta E' = E_2 - E_1$

$$= \frac{-13.6\times 4}{22}\left(+\frac{13.6\times 4}{1}\right)$$

$$\Delta E' = 40.8eV$$

$$\Delta E' = \frac{hc}{\lambda_2} = 40.8eV$$

$$\lambda_1 / \lambda_2 = 2.4$$

18. **2**

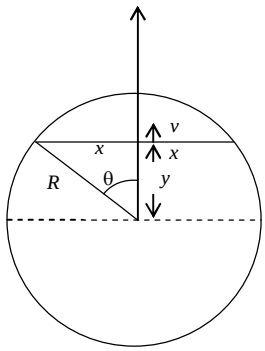
$$F/m = \frac{2ix\mu_0 ni}{m} = \frac{vdv}{dy}$$

Put $\quad x = R\sin\theta; y = R\cos\theta$

$$dy = -(R\sin\theta)(d\theta)$$

Rearrange the variables and Int.; on both sides

$$v = 2ms^{-1}$$



Chemistry

PART – II

SECTION – A

19. **AC**

The volume is $\bar{a}\cdot(\bar{b}\times\bar{c})$. Take $\bar{a}=a\cdot i$

Now $\bar{b}=(a\cos\theta)\bar{i}+(a\sin\theta)\bar{j}=\frac{a}{2}\bar{i}+\frac{a\sqrt{3}}{2}\bar{j}$

$\bar{C}=\frac{a}{2}\bar{i}+\frac{a}{2\sqrt{3}}\bar{j}+\frac{a\sqrt{2}}{3}\bar{k}$

$\therefore \bar{a}\cdot(\bar{b}\times\bar{c})$

$$=\begin{vmatrix} a & 0 & 0 \\ \frac{a}{2} & \frac{a\sqrt{3}}{2} & 0 \\ \frac{a}{2} & \frac{a}{2\sqrt{3}} & \frac{a\sqrt{2}}{\sqrt{3}} \end{vmatrix} = a^3\times\frac{\sqrt{3}}{2}\times\frac{\sqrt{2}}{\sqrt{3}}=\frac{a^3}{\sqrt{2}}$$

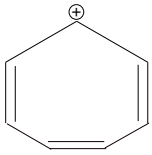
20. **AB**

Due to lanthanide contraction size of $\mathrm{Zr}\simeq\mathrm{Hf},\mathrm{Nb}\simeq\mathrm{Ta}$

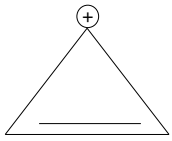
21. **BCD**

(A) Non - aromatic

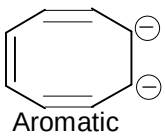
(B)



(C)



(D)



Aromatic

22. **ACD**

23. **AC**

On decreasing pressure concentration decreases so equilibrium shift in forward direction so degree of dissociation increases but in net concentration of all decreases

Molecular wt. of PCl_5 = (1 + α) Molecular wt. of mixture

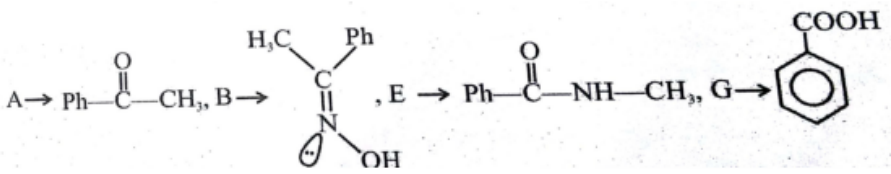
$M_{PCl_5}=(1+\alpha).2.V.D_{mix.}$

$V.D\,\mathrm{mi}\,x.=\frac{M_{PCl_5}}{2(1+\alpha)}$

$\alpha\uparrow\ so, V.D\,\mathrm{mi}\,x\downarrow$

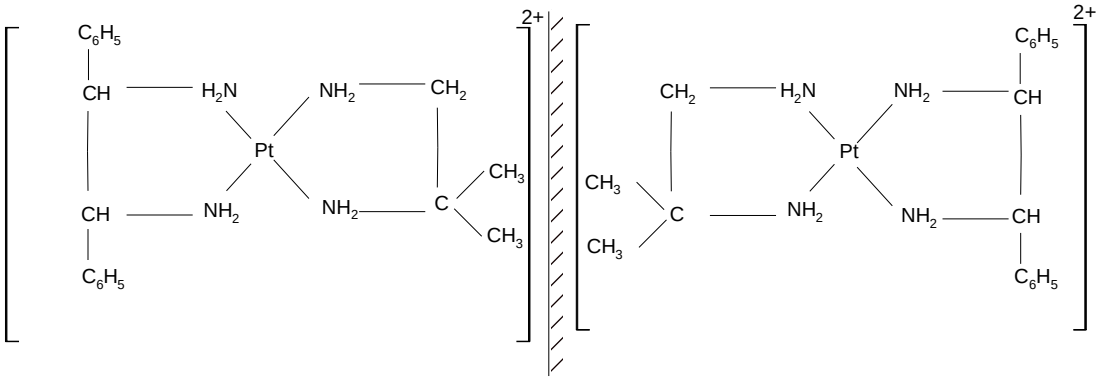
24. **BCD**

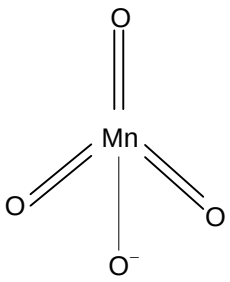
Sol.



25. **ACD**

Square planar complex can show enantiomerism.

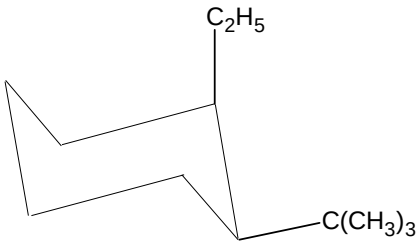




Tetrahaedral, d^3s hybridisation

26. **B**

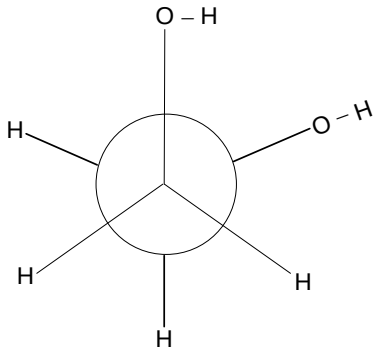
Sol.

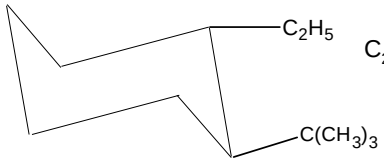


Cis – form

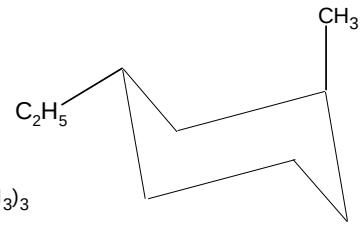
27. **B**

Sol.





Trans – form

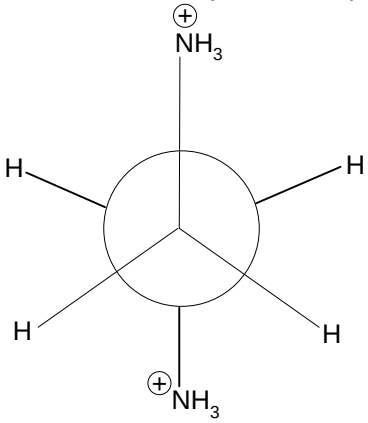


Trans – form

Gauche form stable due to intra molecular H – bonding

28. **C**

In acidic medium pH = 4 in aq. solution



Stable form

29. **B**

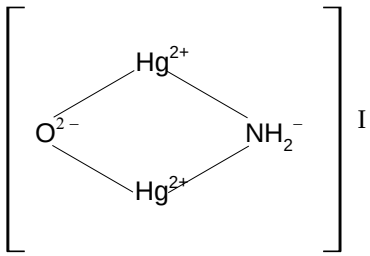
30. **C**

31. **D**

SECTION – C

32. **2**

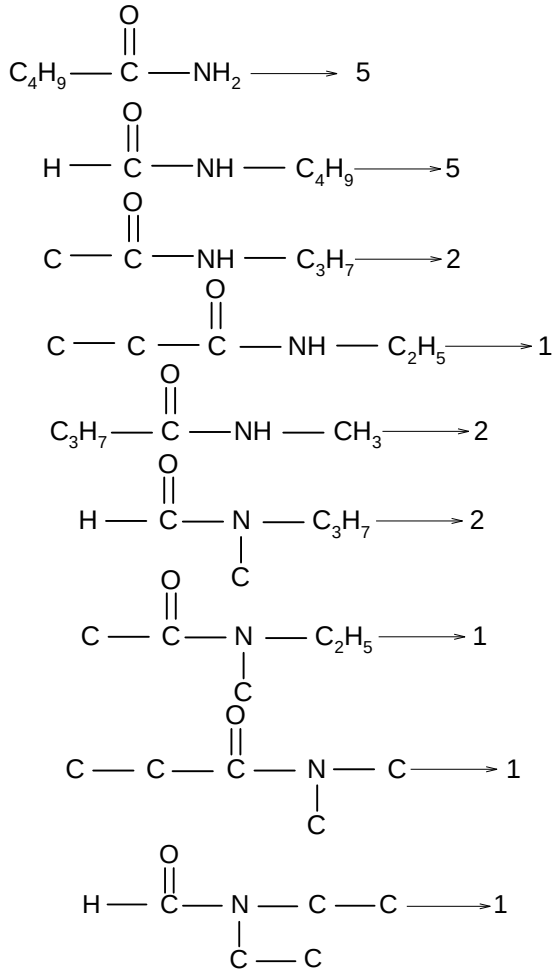
Sol.



Iodide of Millen's base.

33. **5**

Sol.



Total = 20

$X = 20$

$\frac{X}{4} = 5$

34. **4**

$HA \rightleftharpoons H^+ + A^-$

$C_1(1-\alpha_1)$ $C_1\alpha_1$ $C_1\alpha_2$

3.838L volume of water is added so that its conc. will change let C_2. Now new $[H^+] = C_2\alpha_2$

$C_2\alpha_2 = \frac{C_1\alpha_1}{X}$ given

$Ka = \frac{C_1\alpha_1.C_1\alpha_1}{C_1(1-\alpha_1)} = \frac{C_1\alpha_1^2}{1-\alpha_1} = \frac{C_2\alpha_2^2}{1-\alpha_2}$

$1-\alpha_1 \simeq 1$ but $1-\alpha_2 \neq 1$ because on dilution Degree of ionisation increase $\alpha_2 > \alpha_1$

$C_1\alpha_1^2 = \frac{(C_2\alpha_2)^2}{C_2(1-\alpha_2)}$

$C_1\alpha_1^2 = \frac{\left(\frac{C_1\alpha_1}{X}\right)^2}{C_2 - \frac{C_1\alpha_1}{X}}$

$$C_1\alpha_1^2 \cdot \left(C_2 - \frac{C_1\alpha_1}{X} \right) = \frac{C_1^2\alpha_1^2}{X^2}$$

$$C_2 = \frac{C_1}{X^2}\left[1 + X\alpha_1\right]$$

$$\alpha_1 = \sqrt{\frac{K_a}{C_1}} = \sqrt{\frac{1.6\times10^{-4}}{0.1}} = 4\times10^{-2}$$

$$C_2 = \frac{C_1}{X^2}[1 + 4X.10^{-2}]$$

On solving we get X = 4

35. **4**

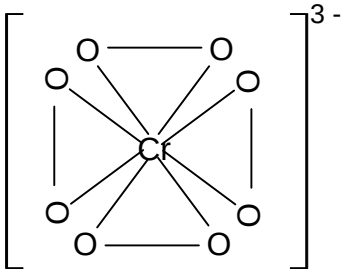
Minimum boiling azeotropes:
Acetone and Benzene
Acetone and carbon disulphide
Acetone and Ethanol
Carbon tetra chloride and Chloroform

36. **6**

$[Fe(H_2O)_5NO]^{2+}$

Oxidation state of Fe = +1 $\Rightarrow$ a

$$K_3[CrO_8] \rightarrow 3K^{1+} + CrO_8^{3-}$$



Oxidation state of Cr is = + 5 = b

$a + b = 6$

Mathematics PART – III

SECTION – A

37. **ABD**

If $\alpha = 0$

$^{n-5}C_0(^{n-5}C_0 + ^{n-5}C_1. + ... + ^{n-5}C_{n-5}) = 2^{n-5}$

$\alpha = 5$

$^5C_5.\left(^{n-5}C_0 + ^{n-5}C_1 + ... + ^{n-5}C_{n-5}\right) = 2^{n-5}$

$\alpha = 1$

$^5C_1.\left(^{n-5}C_0 + ^{n-5}C_1 + ... + ^{n-5}C_{n-5}\right) = 5.2^{n-5}$

$\alpha = 2$

$^5C_2(2^{n-5})$

$\therefore A, B, D$

38. **BC**

$$\int \frac{dx}{x^{1/3}+2} = g(x) + C$$

Put

$x = t^3$

$$g(x) = \frac{3}{2}x^{2/3} - 6x^{1/3} + 12\ln|x^{1/3}+2|$$

$$g(-1) = \frac{15}{2}$$

39. **BCD**

No. ways = $\frac{1}{n+1}\,{}^{2n}C_n$

40. **AB**

$$y = \log_{\frac{1}{3}}(x - 0.\overline{3}) + \log_3\sqrt{(3x-1)^2}$$

$$= -\log_3\left(\frac{3x-1}{3}\right) + \log_3|3x-1|$$

$x > \frac{1}{3}$

$\therefore y = 1$

$x = \pm 2$ $\quad -2$ Rejected

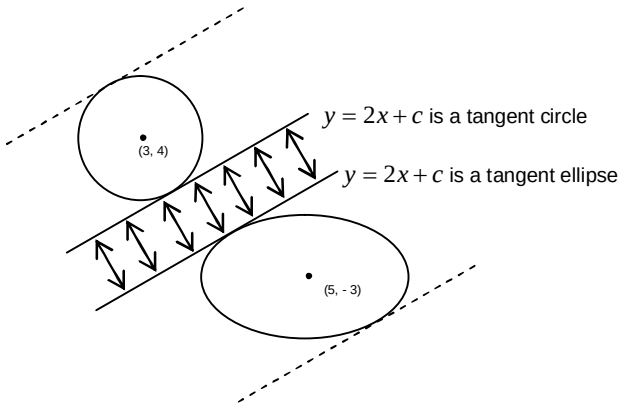
$\therefore x = +2$

$A \equiv (2,1)$ $\quad B \equiv (1,1)$

(A), (B) option satisfy

41. **D**

Sol.



42. **BC**

Points of discontinuities will be

$x = 0,\ x = \frac{1}{2} x = 1, x = 2$

$M = \frac{1}{2}\cos 2t + 1\cos t + 2$

$= \cos^2 t + \cos t + \frac{3}{2}$

$= \left(\cos t + \frac{1}{2}\right)^2 + \frac{5}{4}$

$(B), (C)$

43. **B**

is incorrect

$\angle A = 90^o$

44. **D**

45. **A**

46. **B**

Sol. **(for 44-46)**

(I) $f'(x) > 0$ and Range is a subset of codomain

(II) It is not a function

(III) Decreasing function from $[-1,1]$

(IV) $f'(x) > 0$.

47. **C**

48. **D**

49. **D**

Sol. **(for 47-49)**

(I) Consider $P(x) = ax^4 + bx^3 + cx^2 + dx + e$

$\lim_{x\to 0}\left|\frac{P(-x)}{2x^3}\right|^{\frac{1}{x}}$ must be of the form 1^{∞}

(II) $Q(x) = x - k$ where $k = \int_0^1 f(t)\,dt$

$$f(x) = \frac{x^3}{3} + 1 - x\int_0^x (t-k)\,dt$$

$$= \frac{x^3}{2} + 1 - \frac{x^3}{2} + kx^2$$

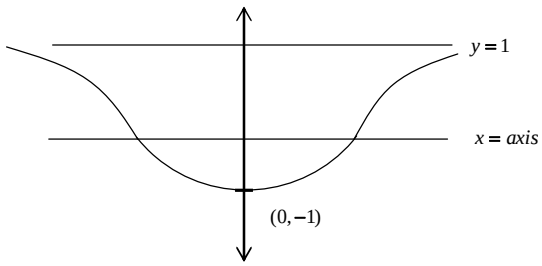
$$= 1 + kx^2$$

$$k = \int_0^1 (1 + kt^2)\,dt = 1 + \frac{k}{3}$$

$$\Rightarrow k = \frac{3}{2}$$

$$\therefore P(x) = 1 + \frac{3}{2}x^2$$

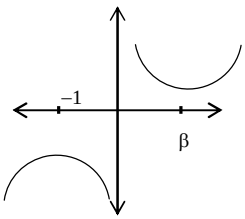
(III)



(IV) $P'(x) = \dfrac{e^{3x}(x+1)(x+e^{-x})(x-e^{-x})}{(xe^x)^2}$

$\therefore x = -1$ Local maxima

$x = \beta$ $\beta \in (0,1)$ local minima



SECTION – C

50. **5**

Perfect squares = x

$1, (2^2, 2^4, 2^6 .....2^{14}), \left(3^2, 3^4, ...3^{10}\right), (5^2, ...5^6)$

$1 + 7 + 5 + 3 + 7\times5 + 5\times3 + 3\times7 + 7\times3\times5 = 192$

Perfect cubes

$1, (2^3, 2^6, ...2^{15})\ (3^3, 3^6, 3^9)\ (5^3 5^6)$

$1 + 5 + 3 + 2 + 5\times3 + 3\times2 + 5\times2 + 5\times3\times2 = 72$

Both

$1, (2^6, 2^{12})\ (3^6)\ (5^6)$

$1+2+1+1+2\times1+1\times1+2\times1+2\times1\times1=12$

Required

192 + 72 – 2(12) = 240

$\therefore$ Answer is 5

51. **2**

n^2+3n+6 is a perfect square

$$n^2+3n+6=y^2$$

D must be 0 or square for n to be an integer

Since

$$n=\frac{-3+\sqrt{9-4(6-y^2)}}{2}$$

$$D=4y^2-15=k^2$$

$$4y^2-k^2=15$$

$$(2y+k)\,(2y-k)=15$$

$$1\times15$$

$$3\times5$$

$\therefore y=4; y=2$

$\Rightarrow n^2+3n-10=0 \qquad n^2+3n+2=0$

$\therefore n=-5,\ n=2,\ n=-1,\ n=-2$

$\therefore\ f(x)=k(x+1)(x+2)(x+5)+x^2$

Since $f(2)=0$

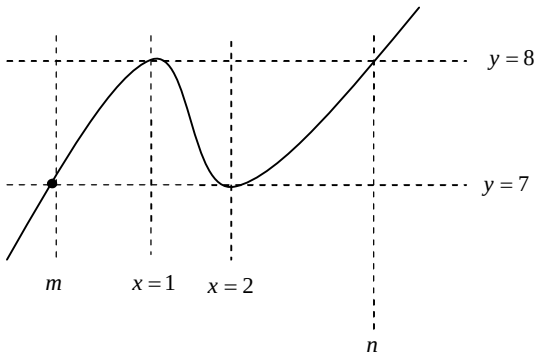
$$k=\frac{-1}{21}$$

Sum of coefficients = $\frac{-5}{7}$

$\therefore$ Answer = 2

52. **3**

Sol. $(-\infty,m)\cup(n,\infty)$



53. **8**

$$\int_0^1 \frac{dx}{\sqrt{1+x}+\sqrt{1-x}+2}$$

Substitute $x=\cos2\theta$

54. **7**

7 points

6 vertices of regular hexagon and one centre of the circle